

TEAM UPDATE : ISC2 x CYBLACK Cybersecurity Bootcamp Recap (May 17-24, 2025)

Hello Team,

Below is a detailed yet concise recap of the sessions of our first week, covering key events, critical cybersecurity concepts, and actionable next steps. This summary provides context for our ongoing learning and highlights why these topics matter in today's cybersecurity landscape.

Context and Purpose

CYBLACK, a charity dedicated to overcoming barriers for Blacks entering cybersecurity, partnered with (ISC)² to deliver this bootcamp. The program offers structured guidance, mentorship, and hands-on learning to prepare us for the (ISC)² Certified in Cybersecurity (CC) certification and beyond. With global cybercrime costs projected to hit \$10.5 trillion by 2025 and 3.5 million unfilled cybersecurity jobs, the bootcamp addresses a critical need for skilled professionals in a rapidly evolving threat landscape.

Key Events and Session Highlights

Saturday, May 17: Onboarding and Cybersecurity Overview

Event: The bootcamp kicked off with an onboarding session led by Dr. Iretoluwa Akerele, CYBLACK co-founder, emphasizing the organization's mission to make cybersecurity education accessible. Alicia Williams from ISC2 introduced the Certified in Cybersecurity (CC) certification.

Key Topics:

- **CC Certification:** Covers five domains (security principles, business continuity, access controls, network security, security operations). It's a 2-hour, 100-question exam requiring a 700/1000 score, with a \$50 annual maintenance fee. Targets students, career changers, and executives.
- **Career Landscape:** Highlighted 10 key fields like penetration testing, cloud security, and trends like AI-powered attacks and zero-trust architecture.
- **Skills for Success:** Foundational IT knowledge, problem-solving, and ethical integrity are critical. A 5-step career entry plan includes learning fundamentals, certification, hands-on experience, networking, and strategic job applications.
- **Why It Matters:** The session set the stage for understanding the certification's value and the vast opportunities in cybersecurity.

Sunday, May 18: Security Concepts and Risk Management

Event: Mr. Roland Mawuli Awuku led a session on the CIA Triad and risk management, with us (participants) showing a 95% comprehension rate.

Key Topics:

CIA Triad: Ensures Confidentiality (restricting data access), Integrity (maintaining data accuracy), and Availability (ensuring system access). Techniques like encryption, hashing, and anti-DDoS protection support these principles.

Authentication and Access: The AAA framework (Authentication, Authorization, Accounting) and non-repudiation ensure secure access and traceability.

Risk Management: Involves identifying, assessing, and mitigating threats. The Equifax breach case study showed how ignoring vulnerabilities led to a massive data breach.

Why It Matters: These foundational principles are the bedrock of cybersecurity, guiding how organizations protect sensitive data and systems.

Wednesday, May 21: Security Controls and Governance

Event: Mr. Roland continued exploring security controls and governance, reinforcing the CIA Triad and risk treatment strategies.

Key Topics:

- **Risk Treatment:** Options include avoidance (eliminating risky activities), mitigation (encryption, training), transfer (insurance), or acceptance (for low-impact risks).
- **Security Controls:** Categorized as preventive (firewalls), detective (intrusion detection), and recovery (backups). The Equifax case underscored the need for timely vulnerability remediation.
- **Asset Protection:** Protecting physical, digital, and intangible assets is critical, with continuous monitoring to adapt to evolving threats.

Why It Matters: Understanding how to prioritize and protect assets helps organizations stay resilient against cyber threats.

Saturday, May 24: Security Controls and Governance (Continued)

Event: The latest session dove deeper into security control mechanisms and governance, emphasizing compliance and incident response.

Key Topics:

- **Control Mechanisms:** Technical (firewalls, antivirus), administrative (policies, training), and physical (locks, access cards). Defense-in-depth stacks multiple controls for robust protection.
- **Security Governance:** Includes policies (acceptable use, data classification), standards (complex passwords), and procedures (phishing response protocols). Compliance with GDPR, HIPAA, and the Ghana/Nigerian Data Protection Act is mandatory.
- **Incident Response:** Procedures for reporting and containing threats (phishing attacks) are critical. A test showed 3/200 employees fell for a phishing link, highlighting ongoing vulnerabilities.

Why It Matters: Governance and compliance ensure organizations avoid legal and financial risks while maintaining trust and operational efficiency.

Key Cybersecurity Takeaways

CIA Triad as a Foundation: Confidentiality, Integrity, and Availability guide all cybersecurity efforts, supported by tools like encryption, digital signatures, and backups.

Risk Management is Ongoing: Continuous identification, assessment, and treatment of risks (internal, external, or multi-party) are vital, as shown by the Equifax breach's catastrophic consequences.

Security Controls are Multifaceted: Combining preventive, detective, and recovery controls across technical, administrative, and physical domains creates a robust defense.

Governance and Compliance: Policies, standards, and procedures ensure legal compliance (e.g., GDPR, HIPAA) and protect sensitive data like PII, safeguarding organizational reputation.

Career Relevance: The growing cybersecurity job market rewards skills like problem-solving and continuous learning, with certifications like ISC2 CC opening doors.

Action Items and Reminders

Stay Engaged: Actively participate, ask questions, and provide feedback. Use full names in sessions for accurate tracking.

Mandatory Attendance: Attend live sessions; recordings are for review only, not primary learning.

Policy Compliance:

- Change passwords every 90 days per company policy.
- Adhere to GDPR for EU data and the Ghanaian/Nigerian Data Protection Act for local data.

Prepare for Next Session: Sunday, May 25 will start our week with Mr. Austin James on Incident Response, Business Continuity (BC) and Disaster Recovery (DR) Concepts/Tasks

Leverage Mentorship: Discuss career-specific questions with assigned mentors for personalized guidance.

Stay Vigilant: Even with training, phishing remains a risk (3/200 employees clicked a test link). Report incidents immediately.

Let's keep the energy high and continue learning together.

Best regards,

Prince Kwesi Lassey (Team Lead) & Joachim Araga (Assistant Team Lead)